

Fault diagnosis and predictive maintenance for hydraulic system based on digital twin model

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Lintao Wang,^{a)}  Yuchong Liu,^{b)} Hang Yin,^{c)} and Wei Sun^{d)}

AFFILIATIONS

School of Mechanical Engineering, Dalian University of Technology, Dalian, China

^{a)} Author to whom correspondence should be addressed: wlt@dlut.edu.cn

^{b)} lyccc@mail.dlut.edu.cn

^{c)} 32004085@mail.dlut.edu.cn

^{d)} sunwei@dlut.edu.cn

ABSTRACT

Hydraulic system has been the mainstream choice in large engineering equipment due to its smooth transmission, large bearing capacity, and small volume. However, because of the tightness and invisibility in hydraulic equipment, it is difficult to check and predict its faults. Common fault diagnosis and maintenance methods for the hydraulic system can be divided into two types: a signal analysis based on the mathematical model and a machine learning algorithm based on artificial intelligence. The first method can only diagnose specific faults based on the mathematical model, which is not universal, and the second one must rely on abundant history fault data, which is impossible to obtain in the early running stage. In order to address these questions, a digital twin framework is proposed which combines the virtual model with the real part to solve practical problems. As a concrete realization form of a five-dimension digital twin model, this framework provides a more feasible solution mode for fault diagnosis in the hydraulic system. Meanwhile, it expands the functions of faults prediction and digital model display. A case study of a hydraulic cylinder is used to illustrate the effectiveness of the proposed framework. The experimental result shows that this method can improve diagnosis accuracy for a hydraulic cylinder greatly compared with the non-interactive simulation model. Meanwhile, with the supplement of actual fault data, the diagnosis accuracy can be further improved, which has a certain growth ability and good applicability.

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I. INTRODUCTION

As the prime transmission means used in large engineering equipment, the digitization and intelligence of the hydraulic transmission have become the basic link to realize the intelligent architecture in the industry. On account of the tightness and invisibility of the hydraulic components, the faults that happened in the devices are not easy to check and remove.^{1,2} Therefore, it is difficult for users to make a reliable diagnosis for fault reasons and assessment for the remaining life of a hydraulic system.

In view of this problem, traditional fault diagnosis methods can be divided into two basic directions: virtual model based and signal processing based.^{3,4} For the situation that it aims to detect several specific hydraulic faults and only has few historical fault data, the effective means is extracting the primary features based on signal processing technology and importing these data to the

corresponding mathematical model for fault diagnosis. Goharri and Sepehri⁵ used the wavelet transform to analyze vibration signals and pressure signals of the hydraulic pump to achieve fault diagnosis. These signal-based methods have been widely applied in the engineering field; however, for a complex system, it is difficult to establish a model according to the coupling between the variables. With the gradual realization of industrial digitization, the application of an artificial intelligence identification method as the main body combined with a variety of feature extraction means has made remarkable progress. Gareev *et al.*⁶ established a simulation model of the helicopter hydraulic system to construct datasets. By comparing two methods including an SVM (Support Vector Machine) classifier based on linear, polynomial, and RBF (Radial Basis Function) kernel and gradient enhancement based on a decision tree, a more suitable machine learning method is chosen for helicopter hydraulic system fault diagnosis. Since the feature data (features

extracted from the output curves of the system) for the training model all come from the initial simulation model, its validity cannot be guaranteed, and the final diagnosis accuracy is relatively low. Ji *et al.*⁷ greatly improved the accuracy of hydraulic valve fault detection by integrating the multiple machine learning model to diagnose results by D-S (Dempster-Shafer) evidence theory and the information fusion method. This method can be applied to almost all basic hydraulic faults, which has high precision; nonetheless, they depend on abundant and effective history fault data, which is always impractical in actual situations. According to the analysis above, these primary hydraulic fault detection methods all have some limitations in different circumstances.

To overcome these restrictions, a more intelligent and comprehensive solution framework should be proposed for the maintenance of the hydraulic system. According to different integration levels, the intelligent process in the industry has undergone three stages of upgrading.⁸ First is the virtual model stage that has no connection with the physical entity. The second stage, digital shadow, only has a unidirectional connection between physical entities and virtual models. The latest version, digital twin, has a bidirectional connection between physical entities and virtual models in the device operation. The original concept of the digital twin was proposed by Professor Grieves⁹ in 2003 whose definition includes three parts: physical entity, virtual model, and their connection. Based on this frame, in order to achieve a more comprehensive acquisition of equipment information and expand its application capacity in life-cycle management, fault diagnosis, prediction, and other aspects, Tao *et al.*^{10,11} extended this framework to five dimensions that include a virtual model, a physical entity, an application function, twin data, and the connection between the four parts, which is more reasonable for the practical application of digital twin technology in the industry.

In practical applications, the frequency of failure in the hydraulic system is very low and the types of faults are not comprehensive, so it is difficult to generate sufficient historical data for the fault diagnosis model in the early or middle period of the equipment operation. Meanwhile, the fault simulation data totally based on a virtual hydraulic model often deviate from the actual system. Considering the problems above and the gradual realization of industrial digitization, an extensible digital twin platform for the hydraulic system based on the five-dimension digital twin model was built. The platform of a two-way interaction can update the virtual hydraulic system model by the real-time feature data selected from the physical entity, and the virtual part can present the situation of the actual system digitally to users. In addition, the model has a certain self-growth ability. After solving the problem of insufficient fault dataset in the early and middle stages of the running time, with the accumulation of actual fault data in the later stage, the digital twin model features data space that can be expanded to improve the fault detection ability of the hydraulic system step by step.

This study is organized as follows: Sec. II introduces the basic structure of the five-dimension digital twin model and specific realization steps to a hydraulic system. Then, Sec. III takes the most common component in the hydraulic system, the hydraulic cylinder, as an example to compare the faults diagnosis accuracy with a machine learning algorithm that is based on three different datasets under the laboratory condition to verify the effectiveness of the digital twin model. Finally, in Sec. IV, an overview of the entire process

and a summary of advantages adopting the digital twin model for hydraulic system maintenance are presented.

II. DIGITAL TWIN (DT) MODEL OF A HYDRAULIC SYSTEM

A. The DT model

Based on the five-dimension digital twin model proposed by Tao *et al.*, a specific twin model frame for the hydraulic system is proposed in Fig. 1, which includes five basic parts: practical hydraulic system, virtual simulation models, user interface used for outputting diagnosis results and maintenance suggestions, twin data, and their interacted manners. By analyzing the relationship between the actual output features and the corresponding simulation output features, and introducing machine learning algorithms for feature data classification and trend prediction of model parameter curves, the digital twin (DT) model can theoretically realize the functions of fault diagnosis, predictive maintenance, expert advice, and other functions.

B. Concrete implementation steps of the DT model

1. Overview of the DT model process

How to effectively use the feature data of the practical hydraulic system and corresponding simulation model to construct efficient fault diagnosis and prediction algorithm is the core question of the DT model. The workflow of the DT model implementation strategy

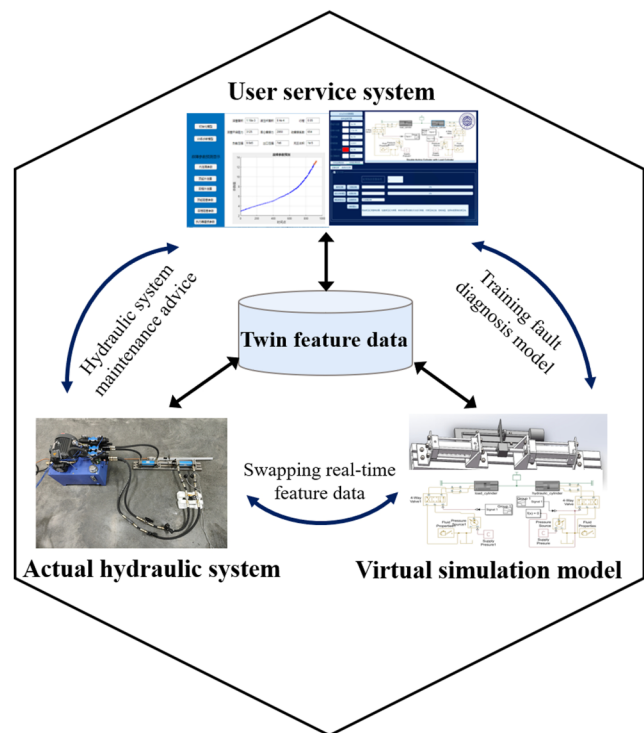


FIG. 1. Basic compositions of a hydraulic system digital twin model.¹¹

for the hydraulic system is shown in Fig. 2, which is described in terms of three main directions:

1. Initial model fitting. In order to minimize the difference between the actual hydraulic system and the corresponding virtual model, the cost functions are constructed depending on their output features. By adjusting basic parameters step by step to decrease the difference between two cost function values to less than 15%, a more realistic simulation model can be achieved.
2. Fault reason diagnosis. After model initialization, the fault modules can be set, respectively, to generate sufficient simulation datasets, which are imported to construct the diagnosis algorithm feature space (projection of feature data in a high-dimensional space). By the real-time feature data, the model can determine the current fault reason. At the same time, the actual fault data and causes are kept in the algorithm feature space to gradually improve the faults diagnosis ability.
3. Predictive maintenance. By analyzing the variation trend of the consumptive parameters with time points in the virtual model, predictive maintenance can be realized. In the actual

operation of the hydraulic system, by collecting the real-time output feature change, the state space model of each fault parameter can be constructed so as to realize the rapid prediction and warning for a practical hydraulic system.

To study the operation in detail, an example is given as follows.

2. Minimum cost function fitting for the virtual hydraulic system

When a new hydraulic system is first deployed, initial device parameters including hydraulic line diameter, line length, and piston radius have been recorded. According to these parameters, a simulation model of the hydraulic system, which is common in the industry, including power components, control components, auxiliary components, hydraulic fluids, actuators, and corresponding loads, can be constructed, as shown in Fig. 3. By adjusting module parameters in the model, different hydraulic systems can be simulated.

However, the equipment parameters used in the model, such as pipe damping coefficient and piston movement damping coefficient, are often not accurate in a different environment. Therefore, the first

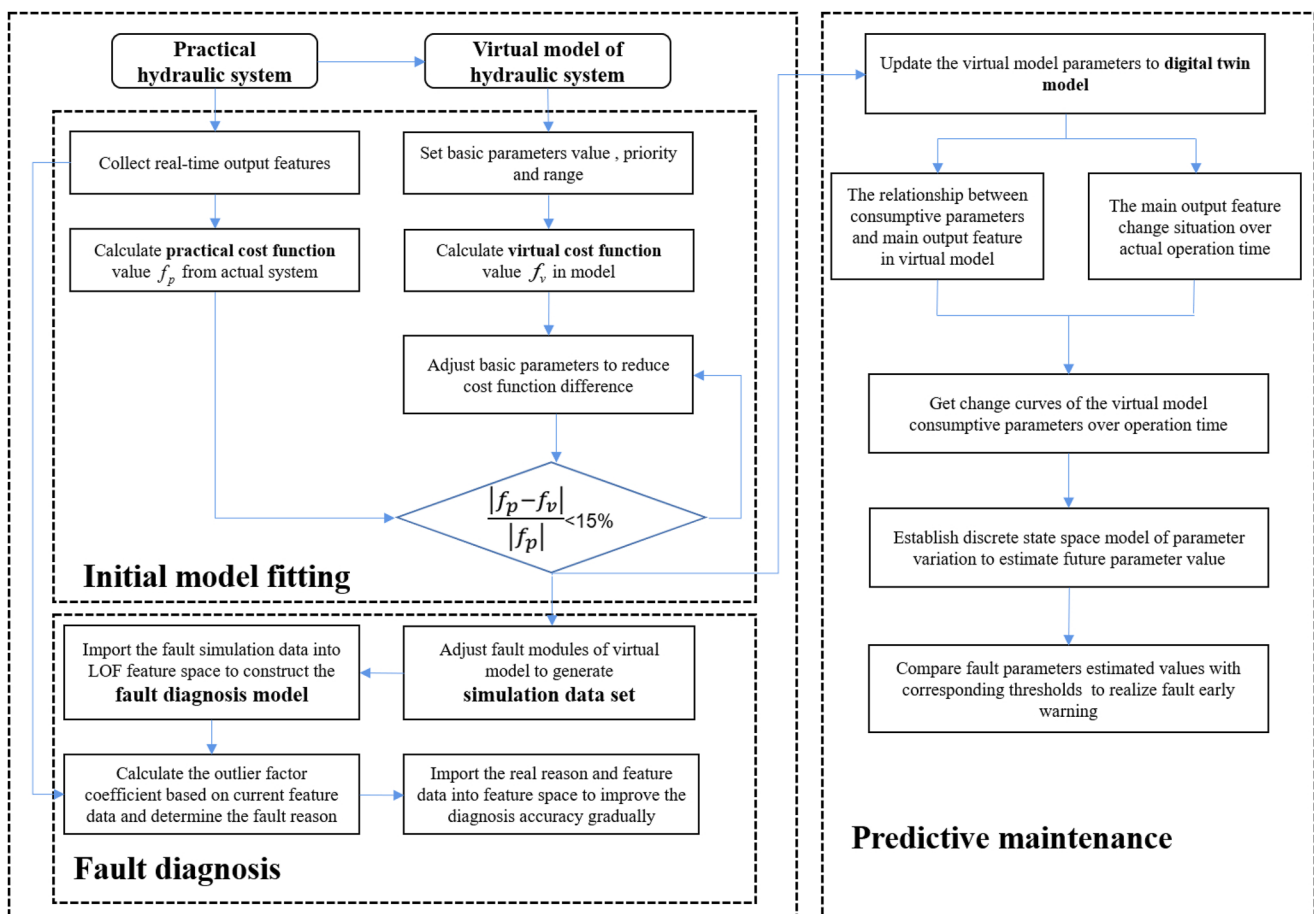


FIG. 2. The basic flow of fault diagnosis and prediction by the digital twin model.

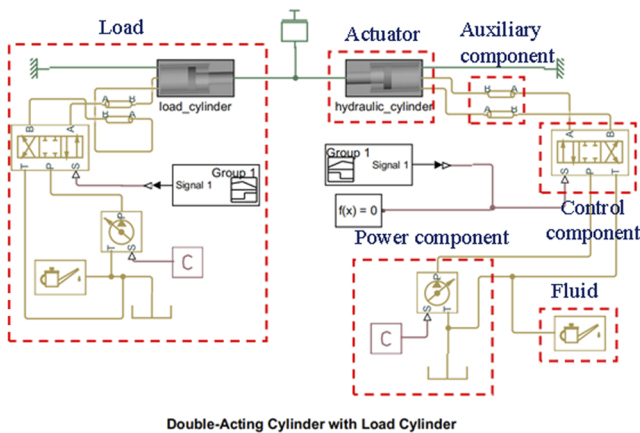


FIG. 3. The hydraulic system model containing six basic modules.

step in constructing a twin model is to correct these parameters in a virtual model, which has only rough ranges in advance. For these parameters, the initial range and priority are set according to experience in the initial case, and the high-priority parameters should be adjusted within the range with a big step to fit the model. Considering the computational cost, the minimum cost function¹² method is adopted. The formula according to the cylinder displacement output features is represented as follows:

$$f_{\text{cost}} = c_1 l_{\text{max}} + c_2 k_{\text{up}} - c_3 k_{\text{down}} + c_4 t_{\text{hold}}, \quad (1)$$

where f_{cost} represents the current cost function value, l_{max} is the maximum distance that the actuator can jack up, k_{up} and k_{down} ,

TABLE I. Formula coefficient value in the virtual and practical hydraulic systems before fitting.

Coefficient name	Coefficient value
c_1	0.2
c_2	0.4
c_3	0.2
c_4	0.2
l_{pmax} (mm)	42
$k_{\text{pup}} (10^{-2} - 100)$	54
$k_{\text{pdown}} (10^{-2} - 100)$	33.3
$t_{\text{phold}} (10^{-1} \text{ s})$	0
f_p	23.6
l_{vmax} (mm)	50
$k_{\text{vip}} (10^{-2} - 100)$	81
$k_{\text{vdown}} (10^{-2} - 100)$	17.5
$t_{\text{vhold}} (10^{-2} - 100)$	4
f_v	39.7

respectively, refer to the absolute slope of the actuator jacking curve and return curve, and t_{hold} stands for the jacking duration. Each parameter used should be adjusted to the same order of magnitude and eliminate the influence of dimensionality. c_1 , c_2 , c_3 , and c_4 are the coefficients whose values depend on the importance of the corresponding output features and the sum is one. The comparison of displacement output curves before fitting is shown in Fig. 4, which shows a large deviation. The example focuses on the fitting of the hydraulic cylinder jacking rate, and the coefficient value and calculation results are given in Table I.

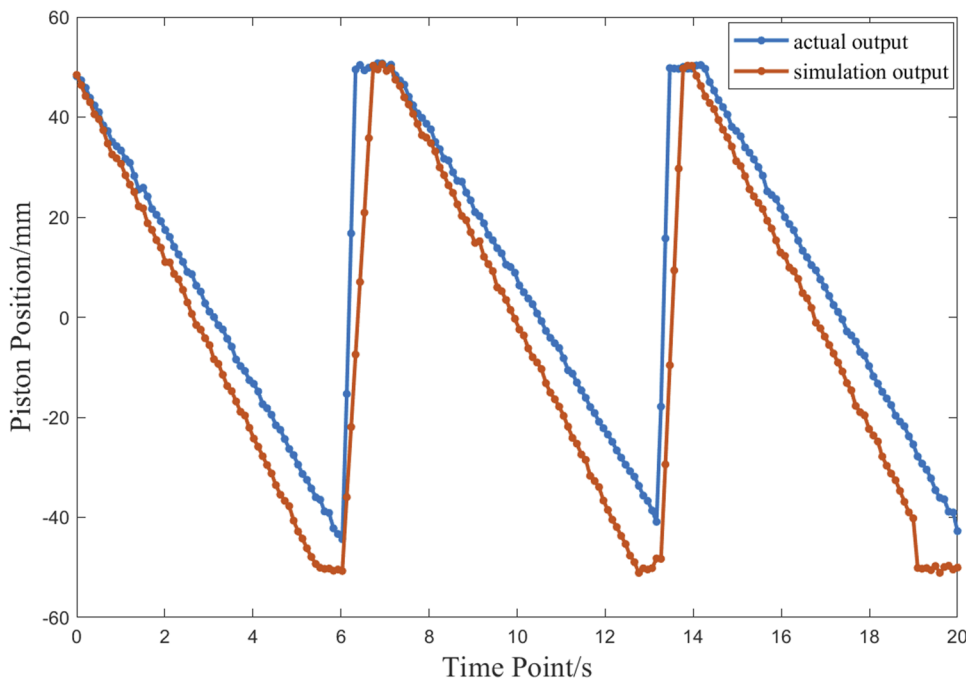


FIG. 4. Virtual and practical output for the piston displacement before fitting.

According to the calculation results, the difference of the displacement output curve before fitting can reach 68%. By minimizing the cost function difference between the actual hydraulic system and the virtual model, these selected parameters are gradually adjusted to a more reasonable value, which is the core to update the virtual model to the DT model. In order to avoid changing only one or two parameters of the model to narrow the cost function gap, a multiple polling method with a small step length is adopted. Under normal operation conditions, the initial virtual model parameters are adjusted on the basis of parameter priority, initial range, and real-time feature data. The change curves are shown in Fig. 5. Six parameters of the virtual model are adjusted step by step to minimize the cost difference value to less than 15%. In this process, the kinetic friction coefficient that has high priority reaches the initial setting range and stops changing before the number of fitting times is 37. Finally, the difference between the simulation cost function and the actual feature cost function is reduced to 6.4%, and the displacement output curves after fitting are shown in Fig. 6, which illustrates a better fit, especially in the jacking speed. The parameters and simulation cost function change results are listed in Table II.

3. Real-time updated fault diagnosis model using LOF algorithm

After solving the initial basic parameter problem of the hydraulic simulation model, a more applicable hydraulic system model can be obtained. By setting various fault modules in the hydraulic system model and gradually adjusting the value of each type of fault parameter, an abundant and effective dataset has been built, which can be used for constructing a related machine learning algorithm to realize fault detection.

Traditional fault diagnosis algorithms, such as support vector machine, K-nearest neighbor algorithm, and integration of multiple machine learning algorithms, have achieved good results with an abundant dataset. However, these methods can only be applied to normal faults, which may cause misjudgment for special faults that are not preset. For the actual feature data newly acquired during the system operation, these commonly used machine learning algorithm models can only be updated through retraining, which will need a high computational cost.

In this paper, a Local Outlier Factor (LOF)¹³ algorithm is used, which can diagnose the special faults by presetting fault threshold and calculating the real-time outlier factor value to decide fault reasons. The outlier threshold value for each fault is 1.5 times the maximum outlier factor coefficient value in the fault simulation feature set, which contains slightly deviated data. If the outlier factor coefficient to every fault set is all greater than the respective threshold, the fault is diagnosed as an abnormal fault, and the actual fault reason is obtained based on practical detection results. In the digital twin model, for this kind of abnormal fault, the feature data collected in real-time and the actual fault reason can be imported into the feature space of the LOF algorithm so as to update the fault diagnosis model to a more comprehensive one.

4. Predictive maintenance based on virtual model fault parameters change

As the hydraulic equipment runs for a certain period of time, many consumptive parameters, such as the blocking coefficient of the pipeline, the sliding friction coefficient of the hydraulic actuator, and inner seal leak area size, will change gradually, which will eventually lead to a large part of failures. At this time, the hydraulic

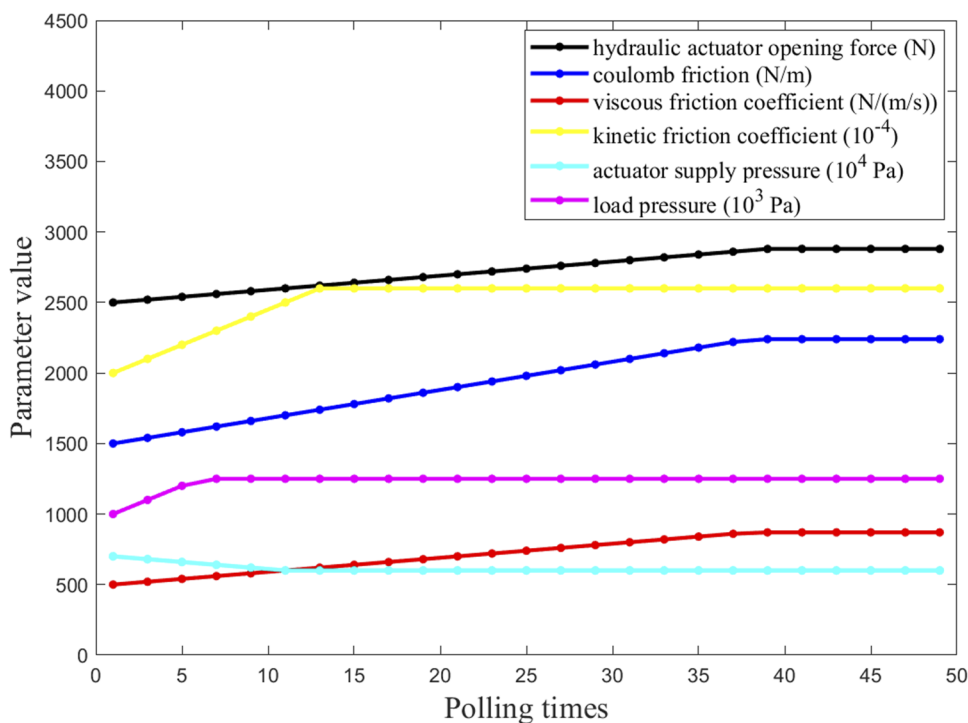


FIG. 5. Correction curve of main parameters using the minimum cost function method.

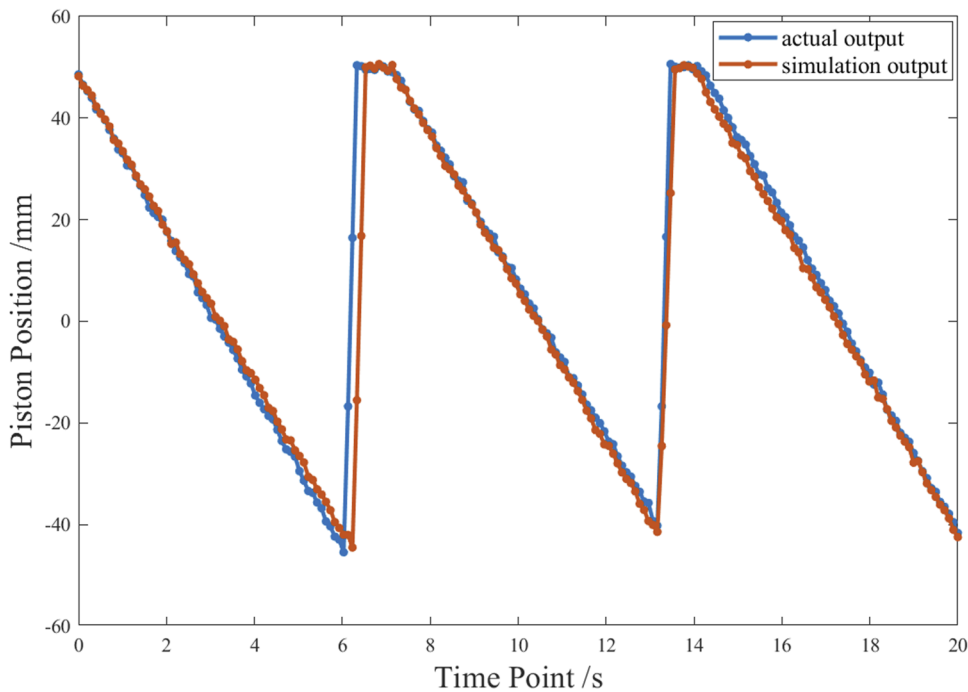


FIG. 6. Virtual and practical output for the piston displacement after fitting.

system is still in the early stage of failure, which has only a slight decrease in the running performance, and it will be difficult to determine these faults that do not occur. Therefore, it is necessary to make an early warning of fault according to the changing trend of performance at this stage. Before common fault prediction, the most likely fault trend in the current actual hydraulic system should be estimated first so as to determine the basic fault parameter that is used for prediction. The LOF algorithm introduced in Sec. II B 3 is adopted here. By calculating the ratio of the outlier factor value to the threshold value for every basic fault, the minimum one is selected. The DT model that has basic fault modules can show the actual hydraulic system changes in real-time, which can facilitate users to have an overall grasp of the actual running situation and realize predictive maintenance.

The total process for the update of model fault parameters is illustrated in Fig. 7. Taking the hydraulic cylinder's inner seal leak parameter change as an example, four main steps for estimation are

introduced. First, the relationship between the main output feature that is extracted from the feature dataset by the principal component analysis method¹⁴ and the inner seal leak parameter is established by running the virtual model and setting the fault parameter step by step. This relation curve shown in (a) part can be described by a cubic polynomial fitting function as follows:

$$f(x) = 0.0439 \cdot x^3 - 0.3372 \cdot x^2 + 1.6545 \cdot x + 1.6651. \quad (2)$$

Second, in a practical hydraulic system, the change curve of the main output feature over time can be selected in the running period. The article adjusts the internal leakage failure module in the experimental platform from small to large to simulate the gradual change of the hydraulic cylinder's inner seal leak condition in a long running time, which is shown in (b) part. According to the fitting function and actual feature data change above, the change curve of the inner seal leak parameter over time point can be obtained, as shown in (c) part. For a quick estimation of the fault parameter, the discrete-time state-space model of the inner seal leak parameter over time point is built, which uses state variables to describe a system by a set of first-order differential equations. The state-space model of the inner seal leak parameter change is as follows:

$$x(t+T) = \begin{bmatrix} 0 & 1 \\ 0.3808 & 0.6364 \end{bmatrix} x(t) + \begin{bmatrix} 1.224 \\ 0.9187 \end{bmatrix} e(t), \quad (3)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} x(t) + e(t), \quad (4)$$

where $x(t)$ is the state variable in current time, $y(t)$ is the sample time that represents the output at instant time t , and $e(t)$ is a single

TABLE II. Main parameters change situation in the virtual model.

Parameter name	Initial value	Adjusted value
Opening force (N)	2500	2880
Coulomb friction (N/m)	1500	2240
Viscous friction [N/(m/s)]	500	1240
Kinetic friction coefficient	0.2	0.26
Supply pressure (103 Pa)	7000	6000
Load pressure (103 Pa)	1000	1250
Simulation cost value	39.7	25.1

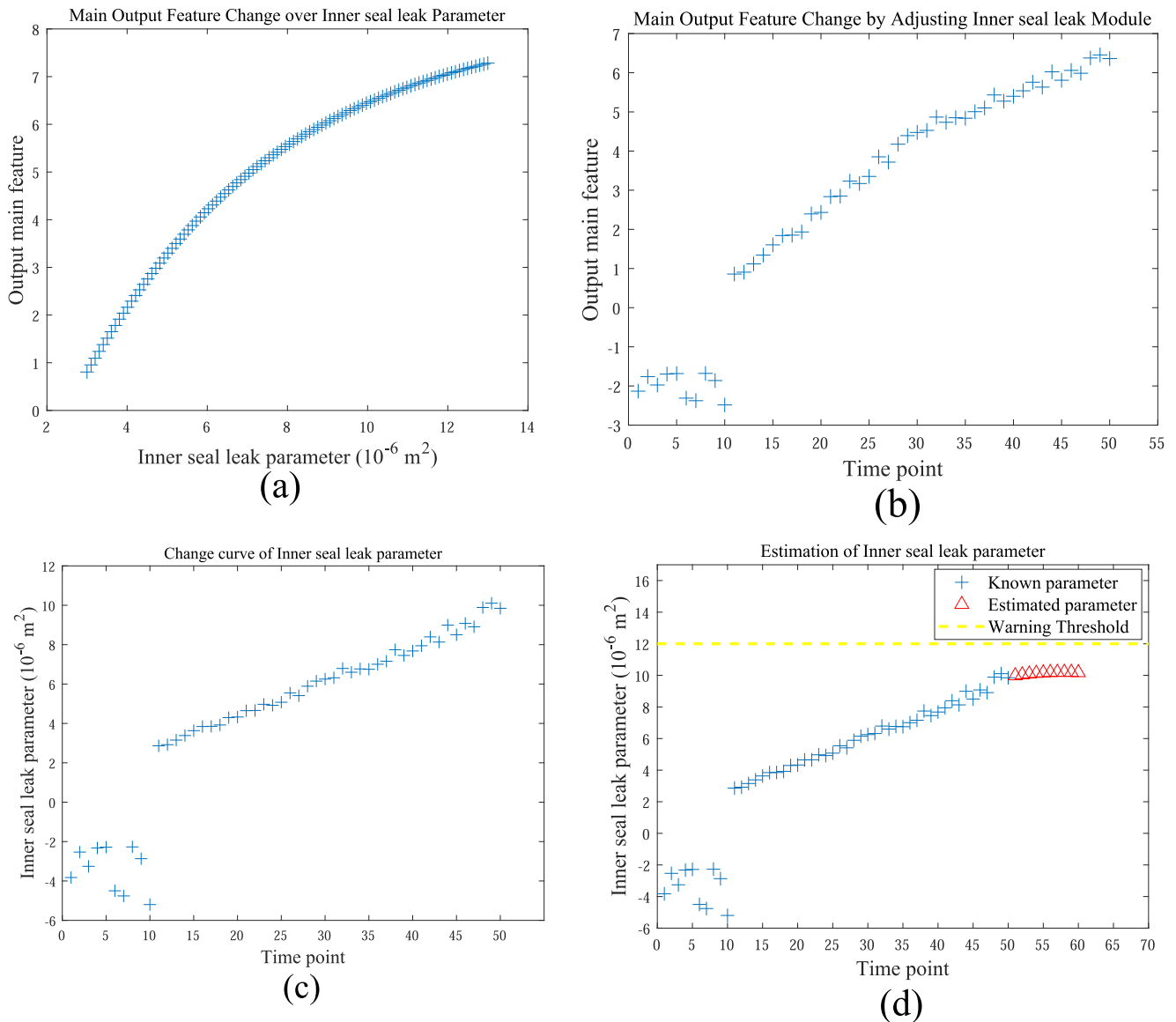


FIG. 7. Four phases in the process of inner seal leak parameter's prediction. (a) Relationship of the main output feature and fault parameter. (b) Main output feature change in the hydraulic system. (c) Fault parameter change over time point. (d) Fault parameter estimation and early warning.

source of noise. The estimation of the parameter is based on this state-space model, and the predicted results are shown in (d) part. In conclusion, by setting a corresponding fault parameter threshold based on experience and comparing the threshold with the current estimated value, the predictive maintenance of the hydraulic system by the DT model can be realized.

Based on the real-time feature data collected from the hydraulic platform and the relationship between the simulation model's output and consumptive parameters, these parameters in the twin model can be updated according to the actual system. After the system runs for a period of time, keep these updated parameters

unchanged and use the initial model fitting method to adjust basic parameters. New fault simulation data can be obtained from the digital twin model, which can be used for training the diagnosis algorithms again and improving their reliability.

III. FAULT DIAGNOSIS COMPARISON RESULTS

In order to verify the diagnosis accuracy and practicability of the DT model in the actual application, a hydraulic cylinder actuator and corresponding pipelines were taken for experimental verification. The experimental platform with common fault modules

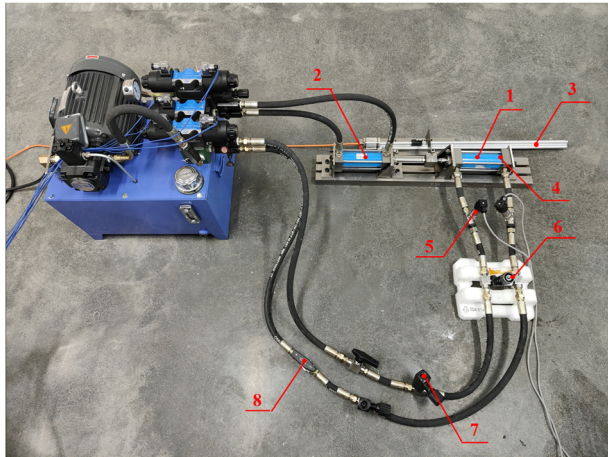


FIG. 8. Hydraulic actuator running platform with faults simulation modules. 1. Hydraulic cylinder actuator. 2. Load hydraulic cylinder. 3. Displacement sensor. 4. Piston wear module. 5. Pressure sensor. 6. Inner seal leak module. 7. External seal leak module. 8. Pipe block module.

is shown in Fig. 8. The test bench was equipped with displacement and pressure sensors and common hydraulic fault modules, such as internal leakage, external leakage, and blocking, can realize the simulation of the common faults and acquisition of actual feature data in different fault situations. Meanwhile, the settings of corresponding common fault modules in the twin model are shown in Fig. 9. The specific fault simulation methods in both the virtual model and actual simulation platform are listed in Table III. Gaussian noise is added to the model output as simulation data.

The simulation data from a traditional virtual model, the feature dataset from the actual hydraulic system, and the simulation dataset from the real-time updated twin model were used to construct the LOF algorithm feature space. To determine the influence of fault diagnosis under three different dataset conditions, 700 groups' practical feature data that tag ten types of basic fault reasons, which include seal leakage in pipes, inner seal leakage, pipe blockage, hydraulic cylinder wear, and their combination, were extracted from this experimental platform by adjusting fault simulation modules. For 70 groups' data collected from the hydraulic platform under each fault situation, if there is a big noise leading to a large

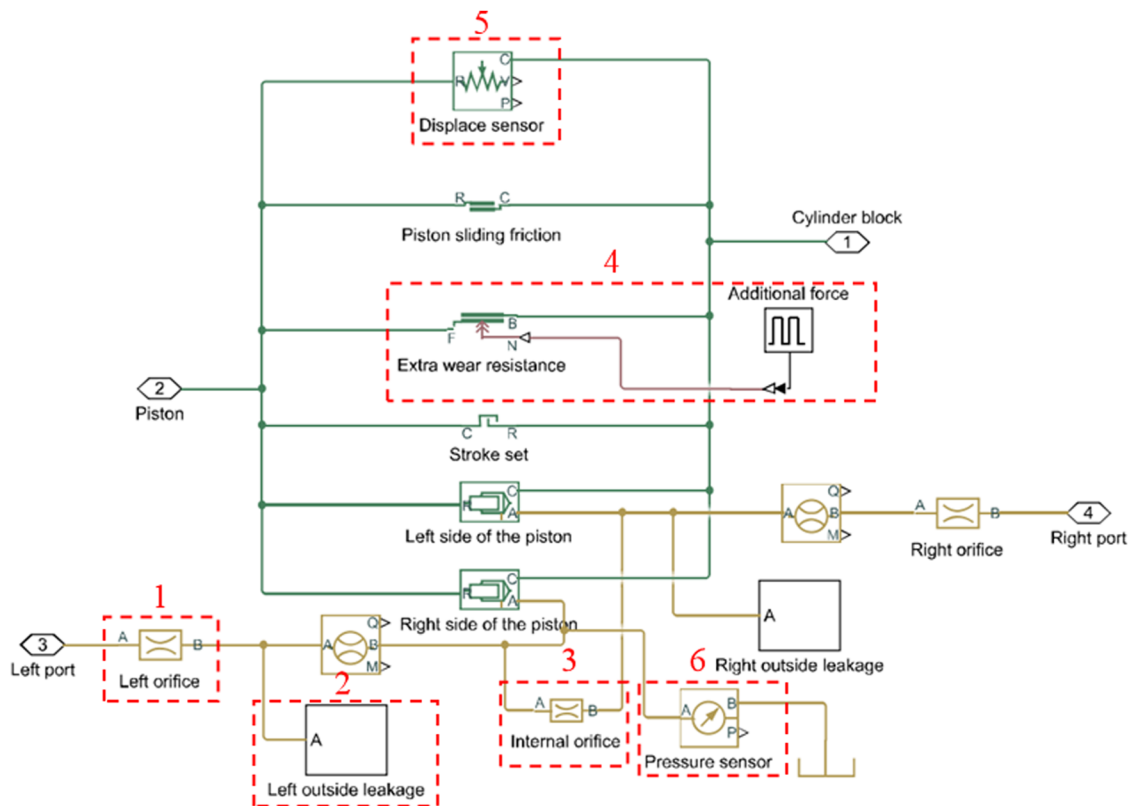


FIG. 9. Hydraulic actuator simulation platform with fault modules. 1. Pipe block module. 2. External seal leak module. 3. Inner seal leak module. 4. Piston wear module. 5. Displacement sensor. 6. Pressure sensor.

TABLE III. The specific fault setting methods in the hydraulic actuator.

Basic fault module	In virtual model	In simulation platform
Pipe block	Connect fixed orifice in pipes	Connect ball valve in pipes
External seal leak	Shunt oil to the tank with throttle valve	Plug in three-way valve and ball valve in pipes
Inner seal leak	Connect inlet and outlet pipes with orifice	Connect inlet and outlet pipes with throttle valve
Piston wear	Add additional force with displacement changes	Change contact surface in piston and cylinder block

deviation in the output curve, the group of data will be directly removed and collected again. In these feature data, 500 groups' data were used as a training set to construct the LOF algorithm, and the other 200 groups, in which each kind of fault has 20 groups, were used as a test set to determine the final diagnosis accuracy. For the two types of simulation data, they are all from the same model with basic fault modules, as shown in Fig. 9. The difference is that the first set of data is directly from the simulation model without a real-time update of parameters, and the second part is from the DT model. In the end, by taking 1.5 times the maximum outlier factor coefficient in each feature set as the threshold, the algorithm's fault diagnosis accuracy for ten faults under three conditions was shown in Fig. 10. The comprehensive diagnosis accuracy of all kinds of fault under the three conditions was listed in Table IV.

According to the experimental results, by setting an appropriate outlier factor threshold, the fault diagnosis accuracy of the

algorithm based on actual fault feature data is the highest, and the comprehensive accuracy can reach more than 95%. However, in the case of the non-interactive virtual simulation model, the diagnosis accuracy is the lowest, and the accuracy in different kinds of fault types fluctuates greatly, which may be caused by the poor fidelity of the simulation model. In contrast, the algorithm trained by the DT model dataset has an obvious improvement and the fault diagnosis accuracy is 89%, which proves that the real-time updated digital twin model did make a difference.

The accuracy of fault diagnosis is the highest by entirely relying on the historical feature dataset, which indicates the importance of the practical data for fault diagnosis and also provides a scheme for improving the ability of the DT model. By putting the real-time feature data that labeled real fault reasons into LOF algorithm feature space that has been constructed by simulation data, the precision of fault diagnosis can be improved gradually.

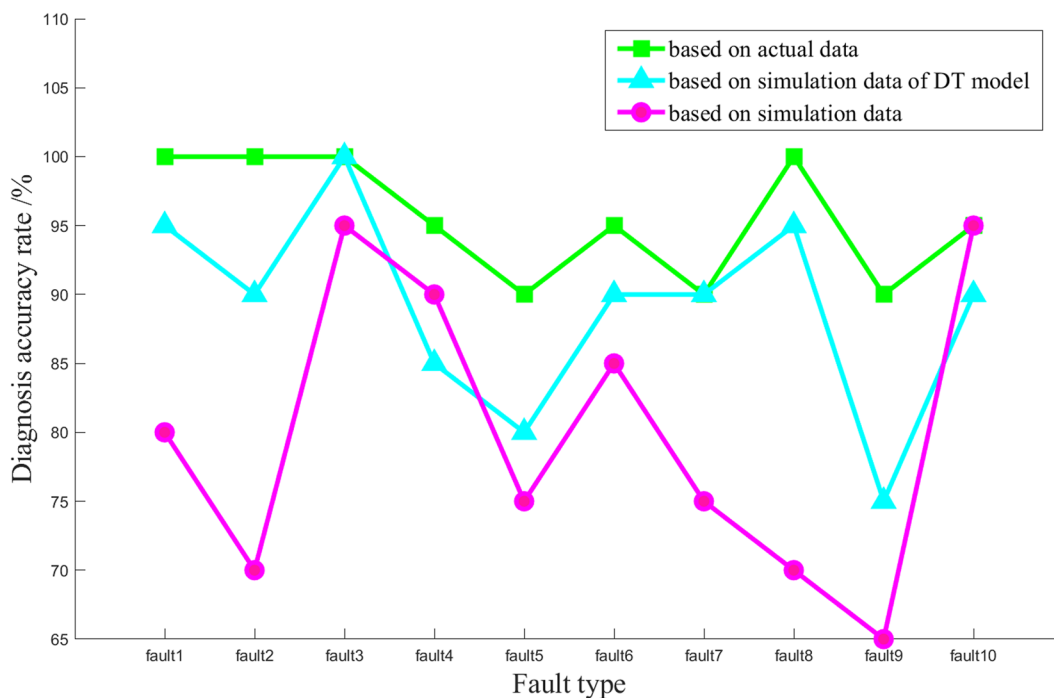


FIG. 10. Diagnosis precision of ten faults based on three models. fault1: external leakage; fault2: pipe block; fault3: inner leakage; fault4: piston wear; fault5: external leakage and pipe block; fault6: external leakage and piston wear; fault7: pipe block and inner leakage; fault8: inner leakage and piston wear; fault9: pipe block and piston wear; and fault10: external leakage and inner leakage.

TABLE IV. Diagnosis accuracy using three different datasets.

Fault dataset	Comprehensive accuracy (%)
Simulation data from virtual model	80
Simulation data from DT model	89
History fault data	95.5

According to the diagnosis results of faults and change trend of common fault parameters, the user interface for the hydraulic system maintenance can be constructed, which is an important part of the five-dimensional model. By setting expert suggestions and warning modes under various fault conditions in advance, users can acquire a maintenance strategy in real-time.

IV. CONCLUSIONS

This paper mainly focuses on the fault diagnosis function of a hydraulic system under different working conditions. A digital twin model based on a five-dimension model is proposed and applied in system prognostic and health management. With the help of the twin model, the inspection and maintenance of a hydraulic system can be extended to the whole life cycle of its operation. Meanwhile, the condition of the system can be displayed to users digitally in real-time. Experimental results show that the diagnosis accuracy of ten common faults of the hydraulic cylinder can reach 89% without the need for sufficient historical fault data, which is an increase of 9% compared with a non-interactive simulation model. Considering the lack of feature data in practical applications, the digital twin scheme with fault parameter prediction and quick model update ability will have significant application value in the future.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Lintao Wang: Conceptualization (lead); Data curation (equal); Investigation (lead); Methodology (equal); Project administration (equal); Resources (lead); Supervision (lead); Writing – review and

editing (lead). **Yuchong Liu:** Data curation (lead); Formal analysis (equal); Investigation (equal); Methodology (lead); Software (lead); Validation (lead); Visualization (lead); Writing – original draft (lead). **Hang Yin:** Conceptualization (equal); Formal analysis (supporting); Methodology (supporting); Validation (equal). **Wei Sun:** Funding acquisition (lead); Resources (supporting); Supervision (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the Dalian University of Technology. Restrictions apply to the availability of these data, which were used under license for this study. Data are available from the authors upon reasonable request and with the permission of the Dalian University of Technology.

REFERENCES

- L. Wang, X. Wu, C. Zhang, and H. Shi, "Hydraulic system fault diagnosis method based on a multi-feature fusion support vector machine," *J. Eng.* **2019**, 215–218.
- D. Huang, L. Ke, X. Chu, L. Zhao, and B. Mi, "Fault diagnosis for the motor drive system of urban transit based on improved Hidden Markov Model," *Microelectron. Reliab.* **82**, 179–189 (2018).
- P. Chen, X. Li, and Z. Tang, "Simulation analysis of steering gear hydraulic system fault mechanism based on AMESim," *2019 5th International Conference on Transportation Information and Safety (ICTIS)*, (2019), pp. 1496–1499.
- J. Dai, J. Tang, S. Huang, and Y. Wang, "Signal-based intelligent hydraulic fault diagnosis methods: Review and prospects," *Chin. J. Mech. Eng.* **32**, 32–75 (2019).
- A. Y. Goharrizi and N. Sepehri, "A wavelet-based approach to internal seal damage diagnosis in hydraulic actuators," *IEEE Trans. Ind. Electron.* **57**, 1755–1763 (2010).
- A. M. Gareev, E. Y. Minaev, D. M. Stadnik *et al.*, "Machine-learning algorithms for helicopter hydraulic faults detection: Model based research," *J. Phys.: Conf. Ser.* **1368**, 052027 (2019).
- X. Ji, Y. Ren, H. Tang, C. Shi, and J. Xiang, "An intelligent fault diagnosis approach based on Dempster-Shafer theory for hydraulic valves," *Measurement* **165**, 108129 (2020).
- W. Kritzing, M. Karner, G. Traar *et al.*, "Digital twin in manufacturing: A categorical literature review and classification," *IFAC-PapersOnLine* **51**, 1016–1022 (2018).
- M. Grieves and J. Vickers, *Trans-Disciplinary Perspectives on Complex Systems* (Springer, Berlin, Germany, 2017), pp. 85–113.
- F. Tao, M. Zhang, Y. Liu *et al.*, "Digital twin driven prognostics and health management for complex equipment," *CIRP Ann.* **67**, 169–172 (2018).
- F. Tao, W. Liu, J. Liu *et al.*, "Digital twin and its potential application exploration," *Comput. Integr. Manuf. Syst.* **24**, 1–18 (2018).
- M. Uddin, S. Mekhilef, and M. Rivera, "Experimental validation of minimum cost function-based model predictive converter control with efficient reference tracking," *IET Power Electron.* **8**, 278–287 (2015).
- H. Ma, Y. Hu, and H. Shi, "Fault detection and identification based on the neighborhood standardized local outlier factor method," *Ind. Eng. Chem. Res.* **52**, 2389–2402 (2013).
- M. Zair, C. Rahmoune, and D. Benazzouz, "Multi-fault diagnosis of rolling bearing using fuzzy entropy of empirical mode decomposition, principal component analysis, and SOM neural network," *Proc. Inst. Mech. Eng., Part C* **233**, 3317–3328 (2019).